

HR Analytics Dashboard

Employee Attrition Analysis using Power BI

Project 3 | Syntecxhub

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1. Introduction

Employee attrition is one of the biggest challenges for any organization, as it increases hiring costs and affects productivity. This project analyses an employee dataset to understand who leaves the company and why. An interactive HR Analytics Dashboard was built in Power BI to help HR teams make data-driven retention decisions.

2. Objectives

- Analyze the employee dataset (salary, department, experience).
- Identify patterns in employee attrition.
- Perform correlation analysis to find key factors.
- Compare attrition across departments and job roles.
- Build KPIs such as attrition rate and retention rate.
- Create an HR dashboard for decision-making.

3. Dataset and Tools

The analysis uses an HR employee dataset of 1,470 employee records covering attributes such as age, gender, department, job role, monthly income, overtime, marital status, total working years and years at the company. A binary Attrition field (Yes/No) shows whether an employee has left.

Tool used: Microsoft Power BI Desktop (data modelling, DAX measures and visualizations).

Derived fields: Age Band, Income Band and Tenure Band were created to group employees for comparison.

4. Key Performance Indicators (KPIs)

KPI	Value	Description
Total Employees	1,470	Total employee records
Attrition Count	237	Employees who left
Attrition Rate	16.12%	Attrition Count / Total Employees
Retention Rate	83.88%	Employees retained / Total Employees

Out of 1,470 employees, 237 have left the organization, giving an attrition rate of 16.12% and a retention rate of 83.88%.

5. Dashboard Walkthrough

5.1 Page 1: Overview

The Overview page gives a quick summary of the workforce. It contains the four KPI cards (Total Employees, Attrition Count, Attrition Rate and Retention Rate), a department-wise attrition chart, a job role-wise attrition chart, Department and Gender slicers for filtering, and a Key Insights panel with recommendations.

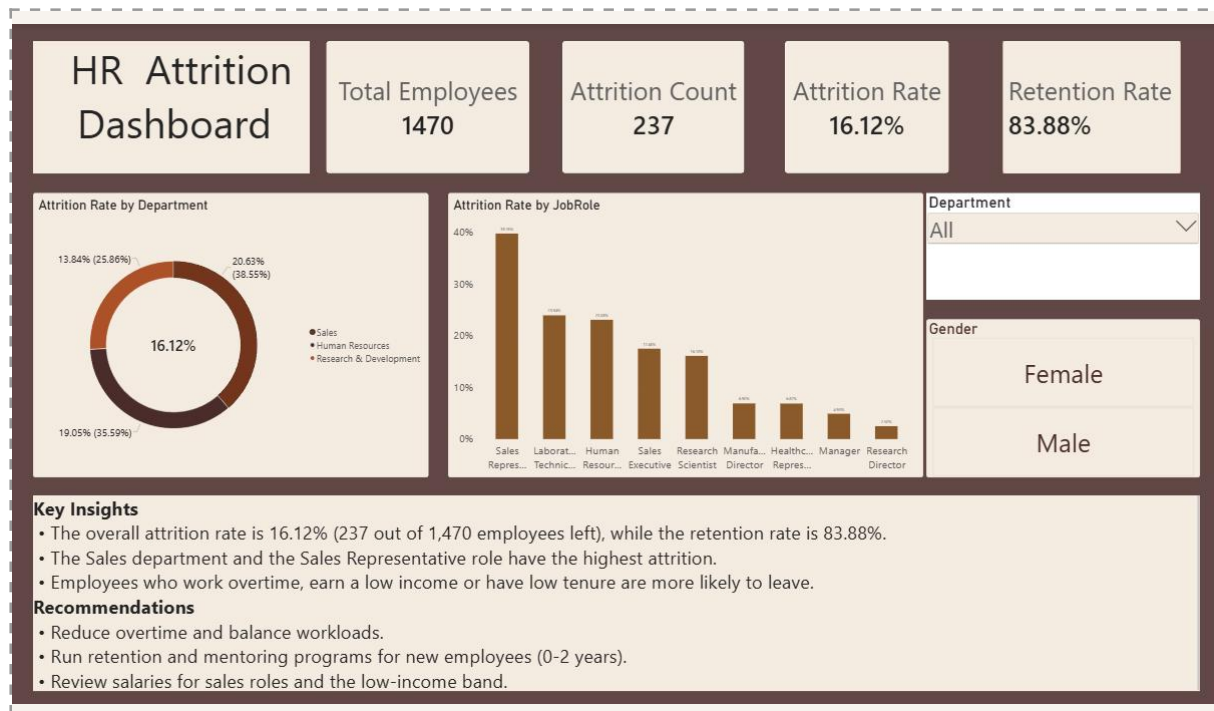


Figure 1: Overview page of the HR Attrition Dashboard

5.2 Page 2: Attrition Analysis

The Attrition Analysis page compares the attrition rate across seven dimensions: Department, OverTime, Marital Status, Age Band, Job Role, Income Band and Tenure Band. This helps pinpoint the employee groups with the highest risk of leaving.

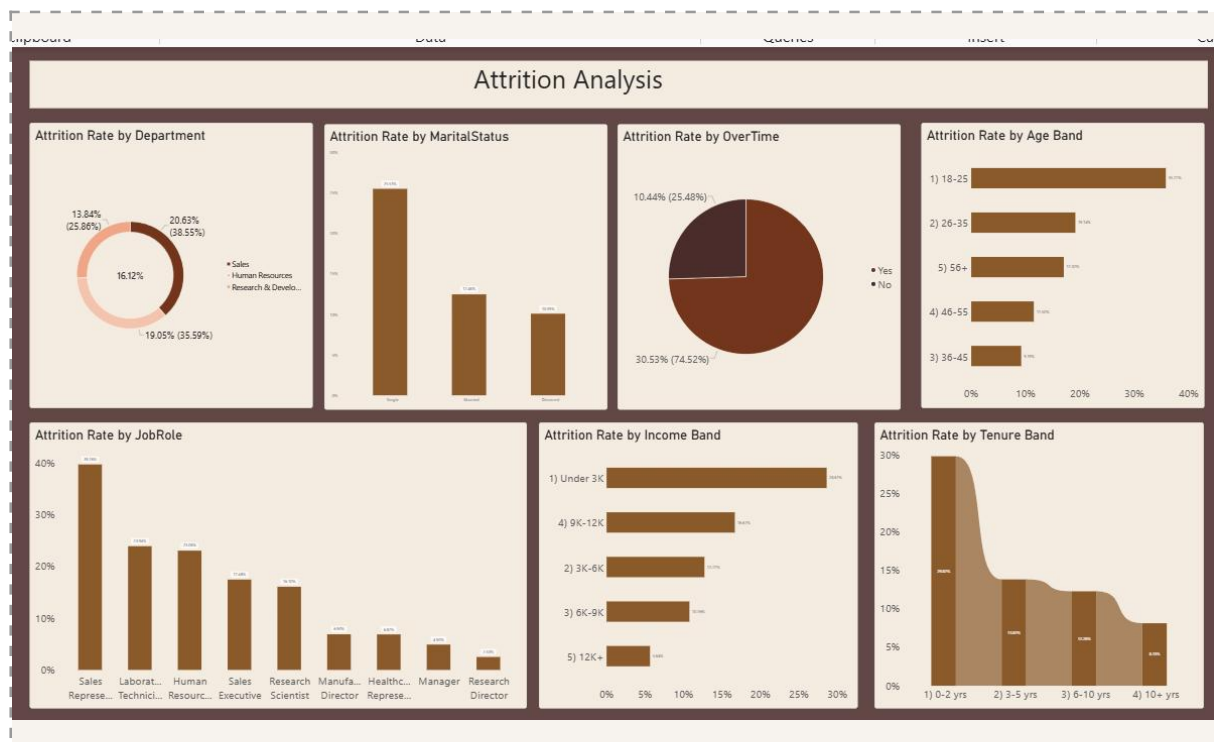


Figure 2: Attrition Analysis page

5.3 Page 3: Correlation

The Correlation page shows how numeric factors relate to attrition. A correlation table lists the coefficients for Age, Distance from Home, Income, Total Working Years and Years at Company. A scatter plot of Monthly Income against Total Working Years, coloured by attrition, shows where employees who left are concentrated.

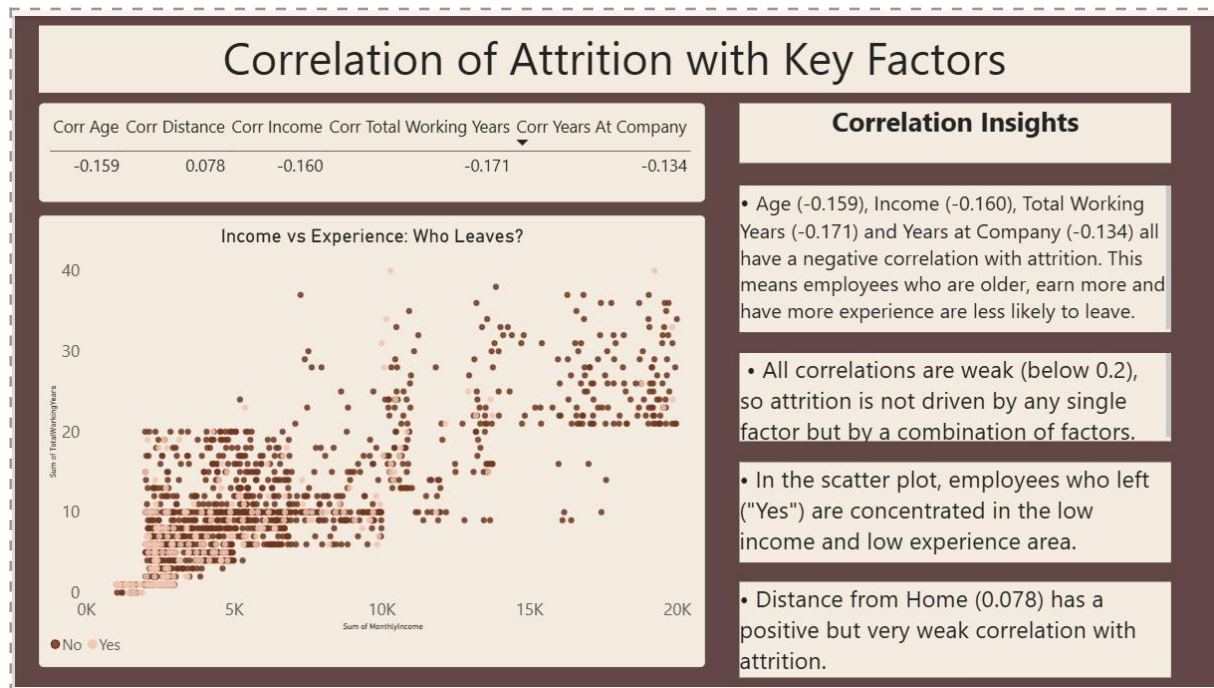


Figure 3: Correlation & 3D page

6. Key Findings

6.1 Department and Job Role

- **Sales department** has the highest attrition rate (20.63%), followed by Human Resources (19.05%) and Research & Development (13.84%).
- **Sales Representatives** have the highest attrition among all job roles, close to 40%. Laboratory Technicians and Human Resources roles also show high attrition, while Managers and Research Directors have the lowest.

6.2 Employee Profile

- **OverTime:** employees who work overtime leave at a much higher rate than those who do not.
- **Age:** the youngest group (18-25) has the highest attrition, and the rate falls as age increases.
- **Income:** attrition is highest in the lowest income band (under 3K) and decreases steadily as income rises.
- **Tenure:** employees in their first two years are the most likely to leave; attrition drops sharply with longer tenure.
- **Marital status:** single employees show higher attrition than married or divorced employees.

6.3 Correlation Analysis

Factor	Correlation	Interpretation
Age	-0.159	Weak negative
Distance from Home	0.078	Very weak positive
Monthly Income	-0.160	Weak negative
Total Working Years	-0.171	Weak negative
Years at Company	-0.134	Weak negative

Employees who are older, earn more and have more experience are less likely to leave. All correlations are weak (below 0.2), so attrition is not driven by one factor but by a combination of factors.

7. Recommendations

- **Reduce overtime:** balance workloads and monitor employees who regularly work overtime.
- **Support new joiners:** introduce onboarding, mentoring and retention programs for employees in their first two years.
- **Review compensation:** revisit salaries for Sales roles and the low-income band to remain competitive.
- **Focus on high-risk groups:** use the dashboard to track Sales, young and single employees more closely.

- **Look beyond one factor:** combine pay, workload, career growth and work environment in retention planning.

8. Conclusion

The HR Analytics Dashboard turns raw employee data into clear insights. With an attrition rate of 16.12%, the analysis shows that attrition is concentrated among employees in Sales roles, those working overtime, and those who are young, low-paid or new to the company. These insights allow HR teams to take targeted action to improve retention and reduce hiring costs.